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Unit → 1

Set - Theory

→ Set → A well define collection of object is called set.

Ex → The collection of all student of AIML 1st year.

Ques → Represent the set of vowels by set builder and roster form?

R_o - {a, e, i, o, u}

Set b = {x : x ∈ Vowels}

Set bu -

Ques → Represent the set of natural number in both form?

Ans → Subset (c)

A = {1, 2, 3, ---- 9}

B ⊂ A

B = {1}

B = {1, 14, 18}

* Union (Common element in all is) →

A ∪ B = {1, 2, 3, 4, 5, 6, 7, 8, 9, 14, 18}

* Intersection (∩)

A ∩ B = {1}

A - B = {2, 3, 4, ---- 9}

B - A = {14, 18}

* Compliment → Let 'U' be the universal set and a set belongs to 'U'. then U - A is called compliment of A 'U'. It is denoted by A^C.

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$U = \{1, 2, 3, ----9\}$
$B = \{7, 8, 9\}$
$B^c = \{1 --- 6\}$

* Symmetric difference →

Let 'U' be a universal set and A and B are two sets then A, B, C U the symmetric difference of A and A ΔB and is defined as A ΔB = (A ∪ B) - (A ∩ B)

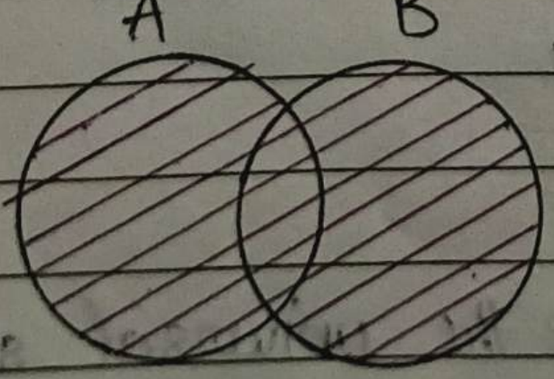
ex→ The symmetric difference of two set is the collection of element that belongs to either of the set but not both.

$A = \{1, 2, 3, 4\}$
$B = \{3, 4, 5, 6\}$

$A \Delta B = (A \cup B) - (A \cap B)$
$= (1, 2, 3, 4, 5, 6) - (3, 4)$
$= \{1, 2, 5, 6\}$

→ Venn diagram → It is dictorial representation of set

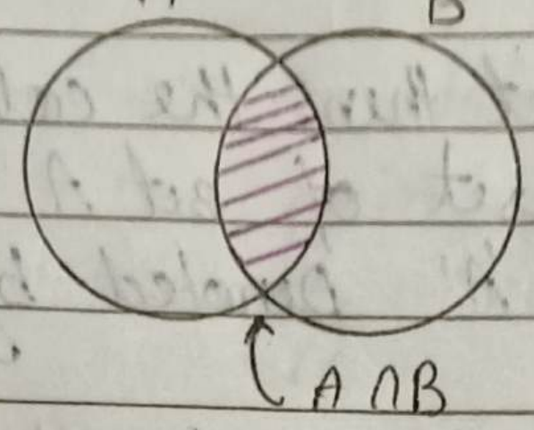
Union →



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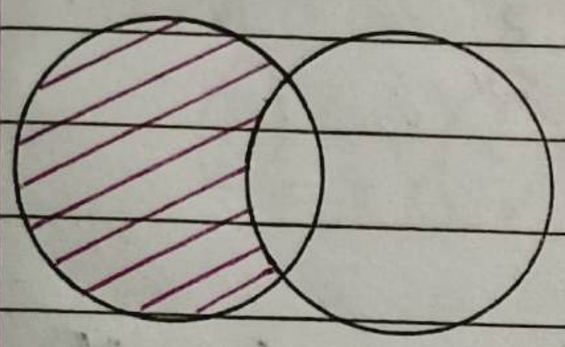
Intersection →

$A \cap B$

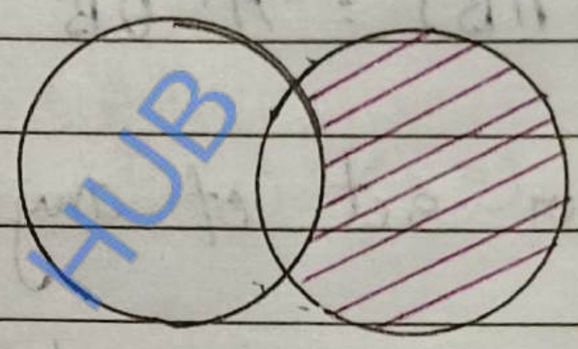


Difference →

A - B

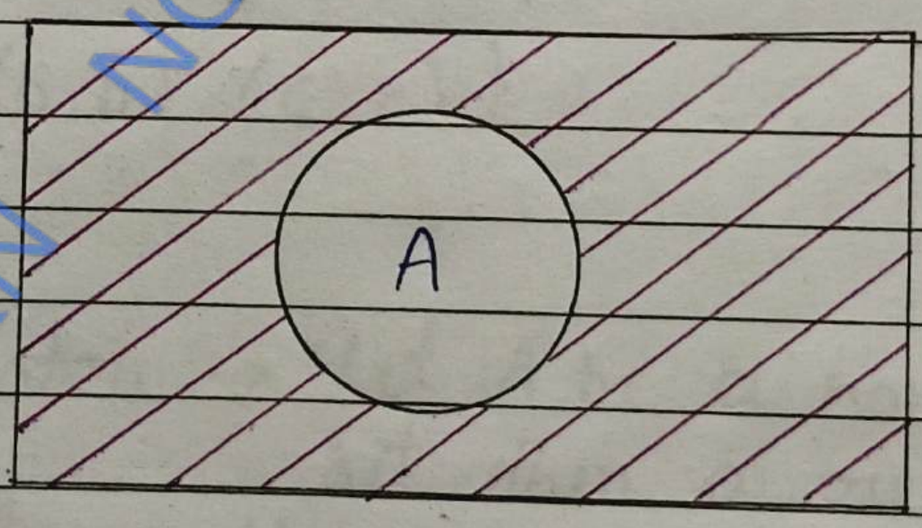


B - A

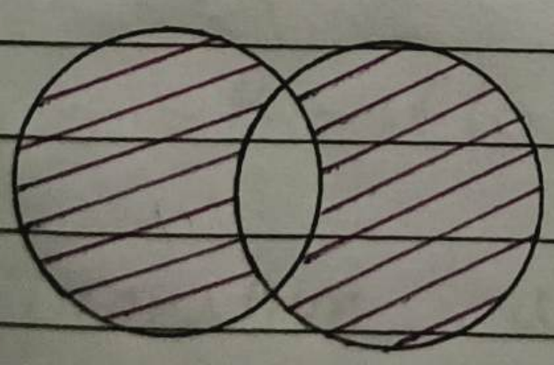


Complement →

AC or A' ←



→ Symmetric difference →



$A \Delta B = (A - B) \cup (B - A)$

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→ B Power set of any set.

Let A be any set then the collection of All possible subyset of set A is called power set of 'A' Denoted by P(A).

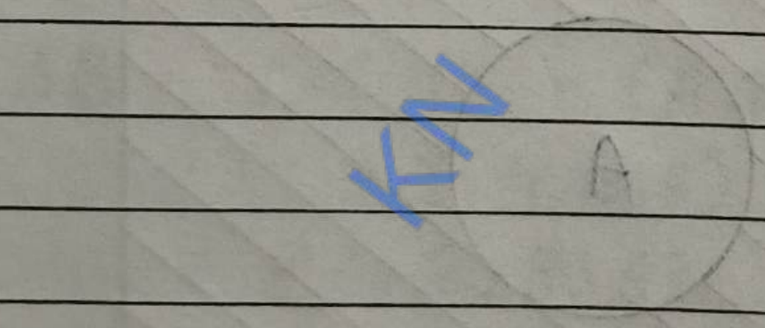
De Morgan's Law / Compliment law. →

$(A \cup B)' = A' \cap B'$
$(A \cap B)' = A' \cup B'$

Power set of any set.

Let A be any set then the collection of all possible subset of set A is called power set of 'A'. Denoted by P(A).

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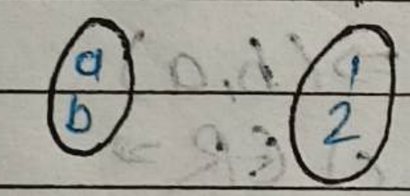


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Types of relation →

a) Empty relation → A relation which contain no pair at all is called empty relation.

b) Universal relation → A relation where every element in a set is paired with every other element.



$R_1 = \phi$

$R_2 = \{(a,1), (a,2), (b,1)(b,2)\}$

i) Reflexive relations :→ In reflexive relation every element is related to itself.

$R_1 = \{(a,a), (b,b), (c,c)\}$

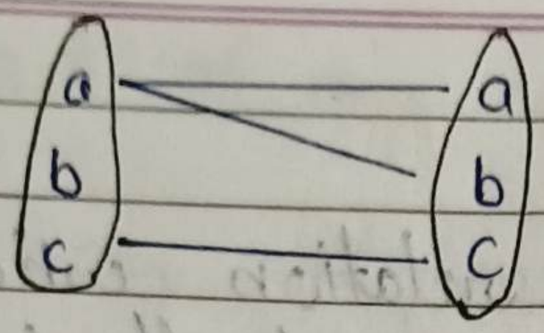
ii) Symmetry relation → Let A be a non-empty set then a relation R defined on A is called symmetric.

if $(a,b) \in R \Rightarrow (b,a) \in R$

$R = \{(a,c), (c,a), (b,b)\}$

iii) Transtive relation - Let A be a non-empty set then a relation R defined on A is called transtive relation.

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$R = \{(a,a)(a,b)(c,c)\}$

iv) Equivalence relation → Let A be a non-empty set then a relation R defined on A is called equivalence if

a) R is reflexive $(a,a) \in R \; \forall \; a \in R$
b) R is symmetric $(a,b) \in R \Rightarrow (b,a)$
c) R is transetive $(a,b)(b,c) \in R \Rightarrow$
$\Rightarrow (a,c) \in R$

Ques → Classify the following relation on set.
$A = \{1,2,3,4\}$

(i) $R_1 = \{(1,1)(2,3)(3,2)\}$
$R_2 = \{(2,3)(4,3)(3,2)(2,2)(4,2)(3,3)(1,1)(4,4)\}$
$R_3 = \{(1,1)(2,2)(3,3)(4,4)\}$

i) Symetric
ii) reflexive, transitive
iii) Equivalance relation

Ques → If R be a relation in the set of the integers defined by $R = \{(x,y) : x \in Z, y \in Z, (x-y) \text{ is divisible by } 2\}$ then prove that R is an equivalence relation?

Given →
$R = \{(x,y):\}$ → Now we have to show that R is reflexive, symmetric and transetive.

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→ R is reflexive → Since 0 is divisible by 2.
⇒ a-0, ---, ---, ---
(a,a) ∈ R ∀ a ∈ Z

R is reflexive →

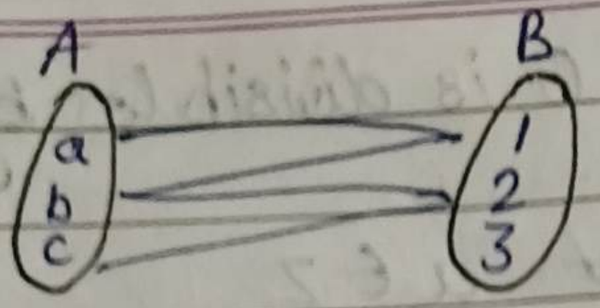
→ R is symmetric - suppose (a,b) ∈ R, Now we have to prove that (b,a) ∈ R

Since → (a,b) ∈ R ⇒ a-b is divisible by 2
= -(a-b) " " 2.
= (b,a) ∈ R

→ R is transitive → suppose (a,b) ∈ R and (b,c) ∈ R
since → (a,b) ∈ R ⇒ a-b is divisible by 2
⇒ a-b = 2k (where k ∈ z) ---①
Now, and (b,c) ∈ R → b-c is divisible by 2
⇒ b-c = 2m (where m ∈ z) ---②
Now, 1+2 eqn: (a-b) + (b-c) = 2k+2m
a-c = 2(k+m)
Since (k+m) is also integer (a-c) is divisible by 2. (a,c) ∈ R
therefore R is transitive]

function →

→ Let a and b be any two sets then 'f' denotes the a rule which associate to each member of 'A' to a member of B then we can say that f is function from A into B where A is said to be domain of function. B is said to be co-domain and image of set A is called range of frame 'f'.



$f(a) = 1$, $f(b) = 1$, $f(c) = 2$

dom $f = A$

Co-dom $f = B$

Range of $f = \{1, 2\}$

Types of function →

1. One-One → A function is said to be one-one if each member of the range of the function arise for the one and only one member of the domain of the function. (injective)

2. many-one → A fun^n $f: A \to B$ is many one fun^n $\exists\ x, y \in A$ such that $x \neq y$ but $f(x) = f(y)$

3. On to fun^n (Subjective) → A function f $A \to B$ is onto fun^n if for each $b \in B$ $\exists\ a \in A$ such that $f(a) = b$

4. Into function → A function $f: A \to B$ is on into function if $\exists\ a \in B$ having no-pre-image in A.

5. Bijective (One One Onto fun^n) → A func^n $f: A \to B$ is bijective fun^n if it is one-one as well onto

Note → If A & B is finite set and $f: A \to B$ is a bijective then A & B have the same number of elements. if

has m elements, then number of bijective from A to B is the total number of arrangement of m items taken all a times i.e. m!

* Identify function → A fun^n $f: A \to B$ is an identify fun^n if $f(x) = x \; \forall \; x \in A$.

* Constant function → A fun^n $f: A \to B$ is constant fun^n if $f(x) = c \; \forall \; x \in A$ where $c \in B$.

Show that →
$f(x) = x^3$ is one one →

suppose $f: \mathbb{R} \to \mathbb{R}$

$$f(x_1) = f(x_2) \quad \forall \quad x_1, x_2 \in \mathbb{R}$$

$$x_1^3 = x_2^3$$

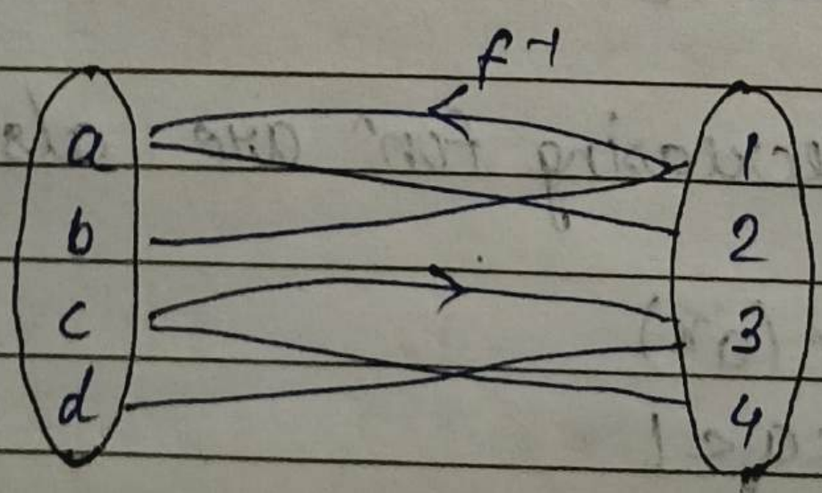
$$x_1^3 - x_2^3 = 0$$

$$(x_1 - x_2)(x_1^2 + x_2^2 + x_1 x_2)$$

$$x_1^2 + x_2^2 + x_1 x_2 \neq 0$$

Invertible function →

→ The function f is set to be invertible function if f^{-1} exist when function f are one-one and onto then inverse of f f^{-1} exist.



$f(a) = 2 \quad f(b) = 1 \quad f(c) = 3 \quad f(d) = 4$

$$f(y^{1/3}) = y$$

$$f^{-1}(y) = y^{1/3}$$

KNOWLEDGE HUB NOTES

Polynomial fun^ :→ f function of the form f(x)

$$a_0x^n + a_1x^{n-1} + \cdots + a_n$$

where $a_0, a_1, a_2 \cdots a_n$ are given real numbers and $a \neq 0$ is called a polynomial fun^ of degree n

Rational fun^ :→ A function of the form →

$$g(x) = \frac{a_0x^n + a_1x^{n-1} + \cdots + a_n}{b_0x^m + b_1x^{m-1} + \cdots + b_m}$$

where → $a_0 \neq 0$, $b_0 \neq 0$

is called rational functional

$$y = \frac{x^3 + x^2}{x^2 + 1}$$

Monotonic function →

A function 'f' is called an increasing function in its domain D if $x_1 < x_2 \Rightarrow f(x_1) \leq f(x_2)$, $\forall\ x_1, x_2 \in D$

Decreasing function →

A function 'f' is called an decreasing function in its domain D if $x_1 < x_2 = f(x_1) \geq f(x_2)$ $\forall\ x_1, x_2 \in D$

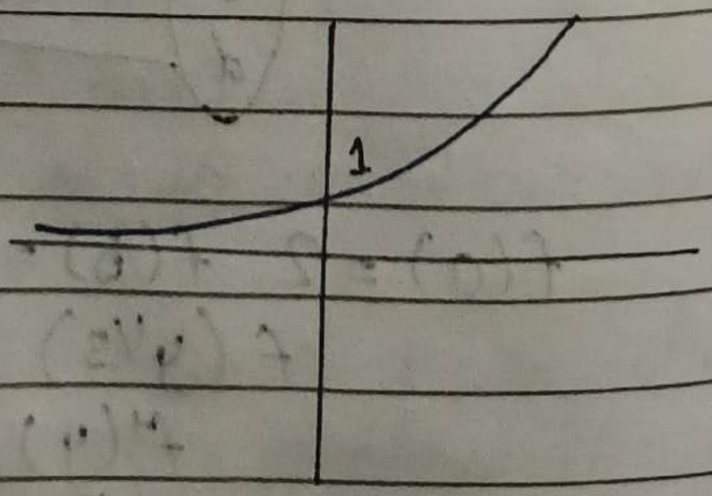
Increasing decreasing fun^ are also called monotonic

Exponential fun^ (ax)

$y = a^x$, $0 < a < 1$

$\lim_{x \to \infty} \left(\frac{1}{2^x}\right) =$

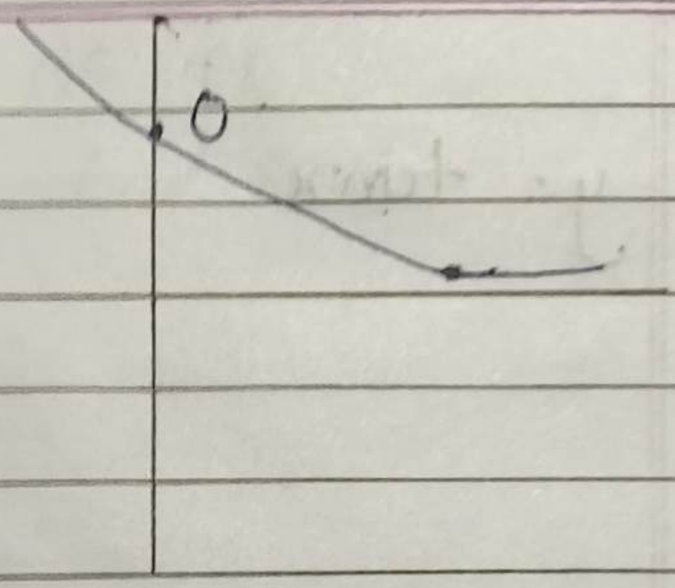
$\lim_{x \to 0} \left(\frac{1}{2^x}\right)^x = 0$



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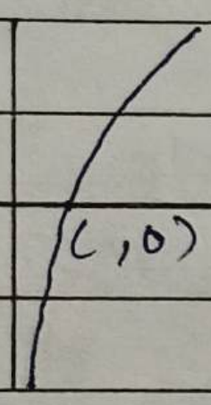
$y = a^x \quad a > 1$

$\lim_{x \to -\infty} \left(\frac{1}{2}\right)^x = 0$

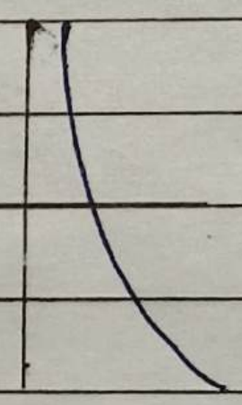


(#) Logrithmic function →

a) $\log_a x \quad a > 1$



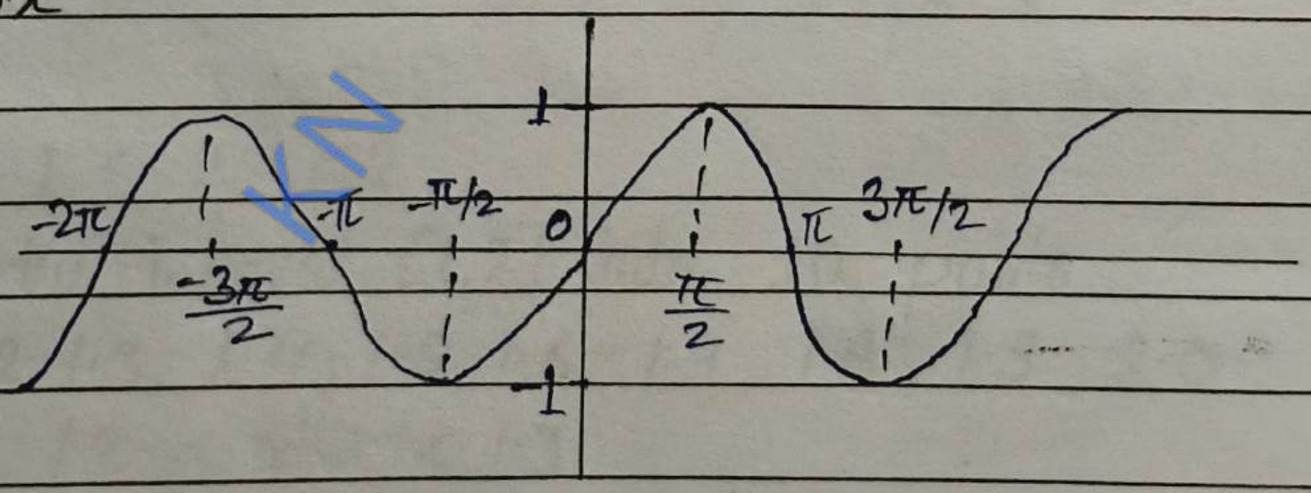
b) $\log_a x \quad 0 < x < 1$



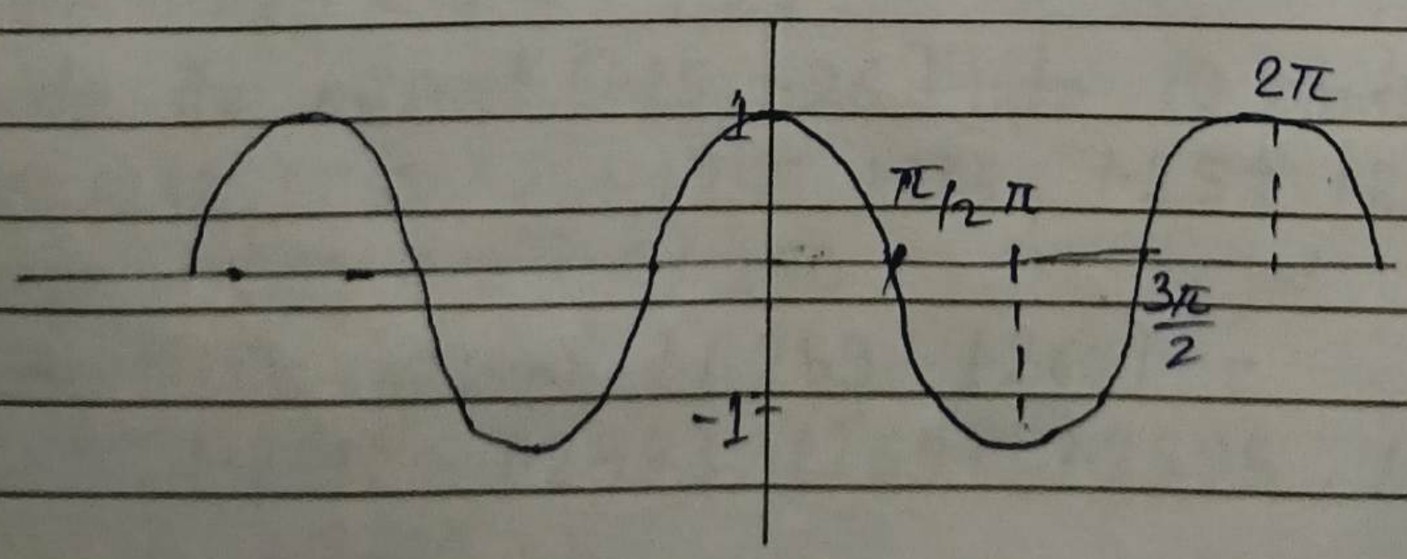
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* Trignometry function →

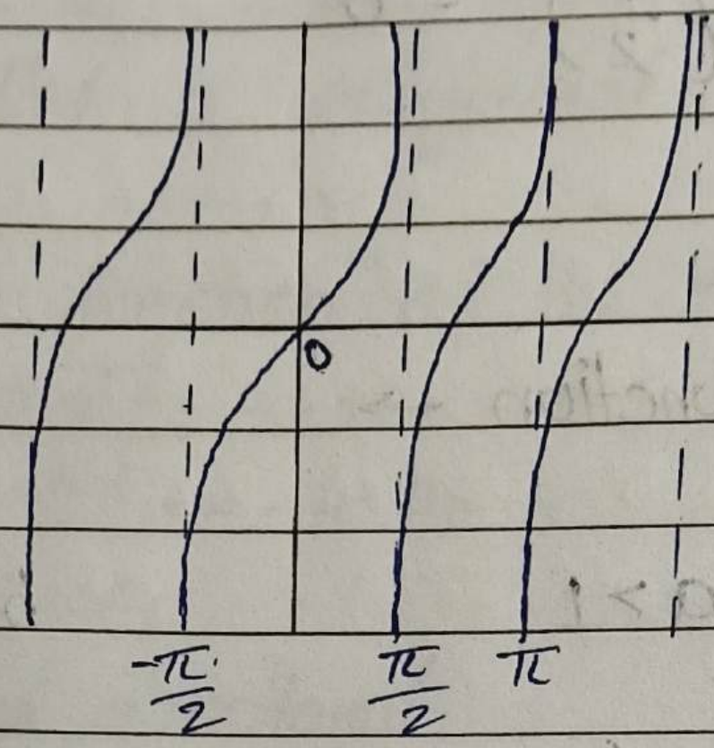
1. $y = \sin x$



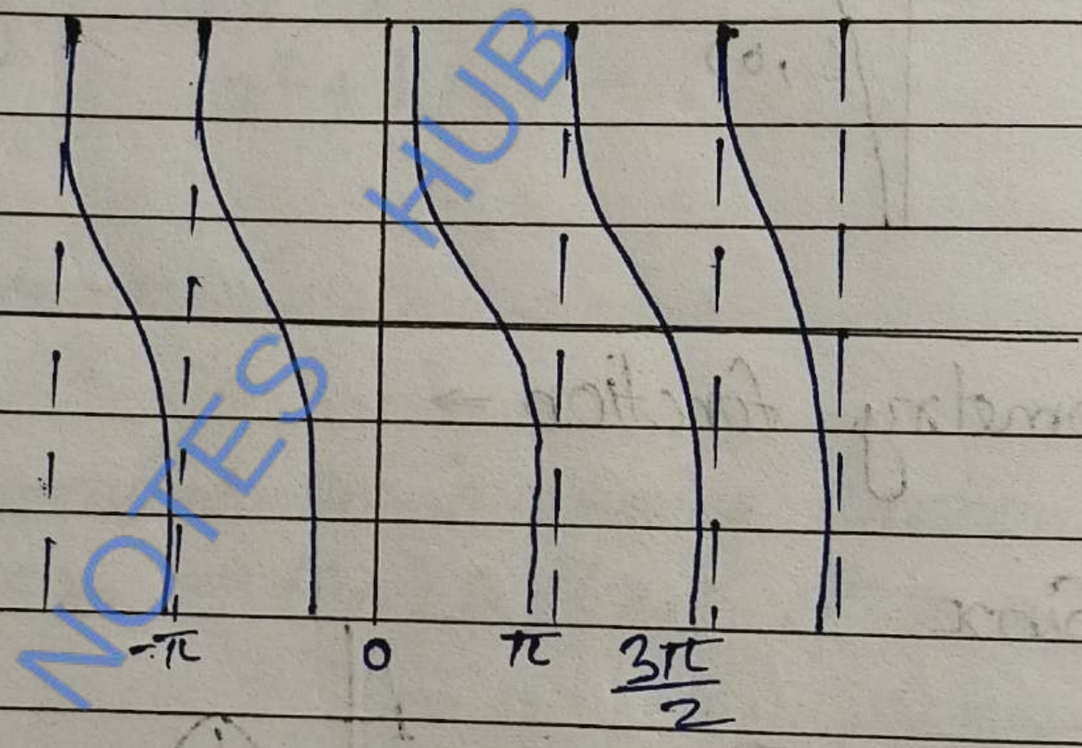
2. $y = \cos x$



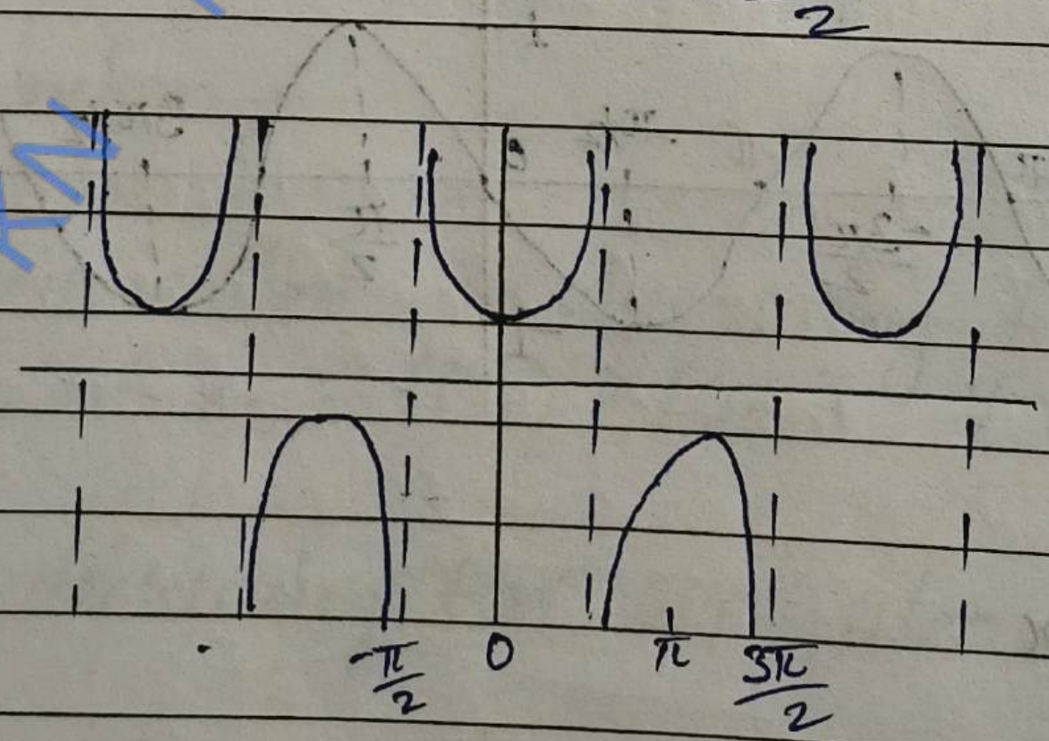
3. $y = \tan x$



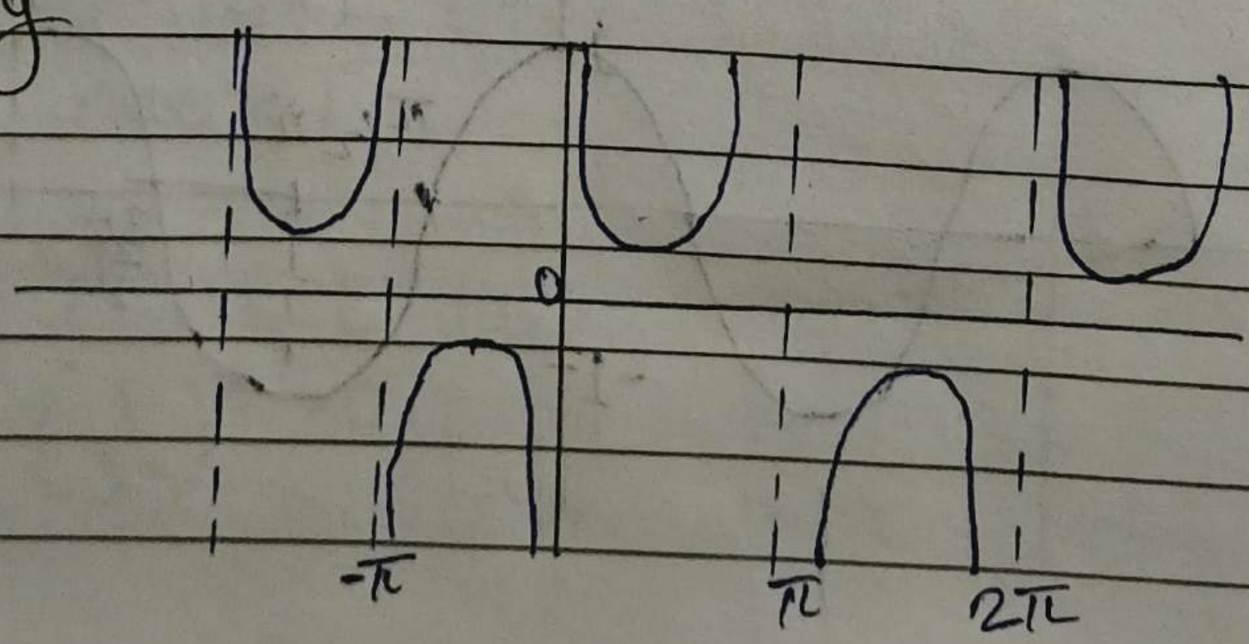
4. $\cot x \Leftarrow y$



5. $\sec x \Leftarrow y$



6. $\operatorname{cosec} x \Leftarrow y$



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whole rational

Number System → $\underset{\text{Natural}}{\mathbb{N}}, \underset{\text{Integer}}{\mathbb{W}}, \underset{\text{Irrational}}{\mathbb{Z}}, \underset{\text{Real}}{\mathbb{Q}}\ \mathbb{Q}^c\ \mathbb{R}$

$$\mathbb{Q} = \left\{ \frac{p}{q} , p, q \in \mathbb{Z} \quad q \neq 0 \right\}$$

2.123
2.12121212 } → $\mathbb{Q}$
2.531694 8 } → $\mathbb{Q}^c$

Ques → Calculate the sq root of 2 up to 3 decimal place.

Since $1 < 2^{1/2} < 2$ $(1 < 3^{1/2} < 2)$
divide the interval [1,2] into 10 parts
1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 2.0
Since [1.7 1.8]
1.71, 1.72, 1.73, 1.74, 1.75, 1.76, 1.77, 1.78, 1.79, 2.80
therefore up to 2 decimal place = 1.73

Ques → Calculate the cube roots of 2 upto 3 - decimal place.

Since $1 < 2^{1/3} < 2$
divide the interval [1,2] into 10 parts.
1.1, 1.2, 1.3, 1.4, 1.5, 1.6 1.7, 1.8, 1.9, 2.0
Since $1.2 < 2^{1/3} < 1.3$
Divide the interval [1.2 1.3] upto 10 parts.
1.21, 1.22, 1.23, 1.24, 1.25, 1.26, 1.27, 1.28, 1.29, 1.30
since $1.25 < 2^{1/3} < 1.26$
Divide the interval [1.25, 1.26] into 10 parts.
1.251, 1.252, 1.253, 1.254, 1.255, 1.256, 1.257, 1.258, 1.259, 1.260
since $1.259 < 2^{1/3} < 1.260$.
Divide the interval [1.259, 1.260]
1.2591, 1.2592, 1.2593, 1.2594, 1.2595, 1.2596, 1.2597
1.2598, 1.2600

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since $1.2599 < 2^{1/3} < 12600$

$2^{1/3} = 1.259$

Ques

The modulus of real no. →

Let x be any given real no. then $|x| = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$

Result

1- $|a+b| \leq |a| + |b|$

2- $|ab| = |a||b|$

3- $|a-b| \geq ||a| - |b||$

Ques Show that $\left|\frac{1}{x}\right| = \frac{1}{|x|}$ if $x \neq 0$

$1 = \frac{x}{x} \quad (x \neq 0)$

$|1| = \left|\frac{x}{x}\right|$

$|1| = |x| \left|\frac{1}{x}\right| \qquad |ab| = |a||b|$

$\frac{1}{|x|} = \left|\frac{1}{x}\right|$

Ques Show that → $\left|\frac{x}{y}\right| = \frac{|x|}{|y|}$, if $y \neq 0$

Let $\frac{x}{y} = z$ then $x = y \cdot z$

taking modulus on both sides

$|x| = |y| \cdot |z|$

$z = \left|\frac{x}{y}\right|$ and divide by $|y|$:

$\left|\frac{x}{y}\right| = \frac{|x|}{|y|}$ (Hence proved)

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Ques if $y = |x| + |x-1|$

$$y = \begin{cases} -x+1-x & x \le 0 \\ x+1-x & 0 < x < 1 \\ x+x-1 & x \ge 1 \end{cases}$$

$$y = \begin{cases} 1-2x & x < 0 \\ 1 & 0 < x < 1 \\ 2x-1 & x \ge 1 \end{cases}$$

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